Time Series Project: IPI Food Industries

Part I: The data

1. What does the chosen series represent?

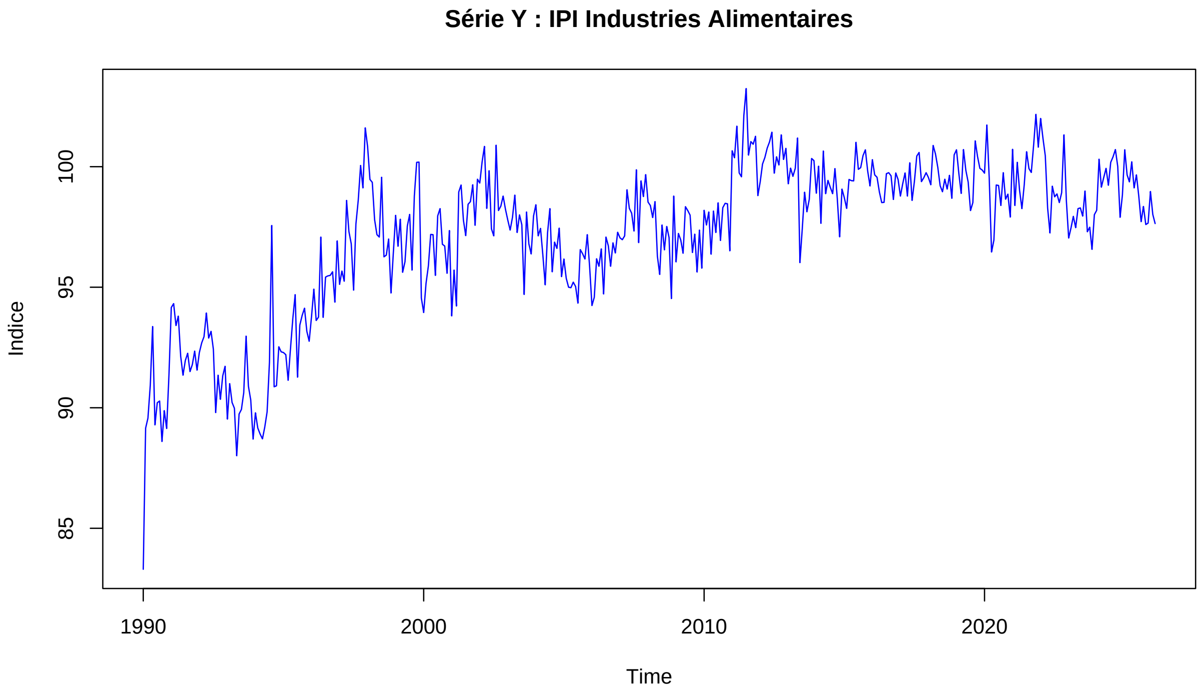
The dataset is an Industrial Production Index (IPI) for the food industries sector in France, with monthly frequency, seasonally and working-day adjusted (CVS-CJO), based at 100 in 2021. The variable is an index and is unit-free. Since the series is already CVS-CJO, we do not apply additional seasonal adjustment.

Modelling context.

- As an index (based at 100 in 2021), values above (below) 100 indicate production above (below) its 2021 average level for the sector.
- The CVS-CJO correction removes most deterministic seasonal patterns and calendar effects; consequently, we do not expect pronounced remaining seasonality and we focus on non-seasonal ARIMA/ARMA dynamics.
- Because the index is strictly positive over the sample, the analysis can be carried out on $Z_t = \log(Y_t)$; first differences of Z_t have a growth-rate interpretation and often provide variance stabilization.

2 and 3. Transform the series to make it stationary if necessary whith graphical representation before and after transformation (placed here for convenience of the study)

```
df <- read.csv("data/valeurs_mensuelles.csv", sep = ";", skip = 4, header = FALSE, stringsAsFactors = FA

df <- df[, 1:2]
colnames(df) <- c("date_brute", "valeur")

df <- df[df$valeur != "" & !is.na(df$valeur), ]

# On ajoute "-01" pour créer un format AAAA-MM-JJ lisible par R
df$date <- as.Date(paste0(df$date_brute, "-01"))
df <- df[order(df$date), ]

# On récupère l'année et le mois de début
start_y <- as.numeric(format(df$date[1], "%Y"))
start_m <- as.numeric(format(df$date[1], "%m"))

Y <- ts(df$valeur, start = c(start_y, start_m), frequency = 12)

plot(Y, main = "Série Y : IPI Industries Alimentaires", ylab = "Indice", col = "blue")
```

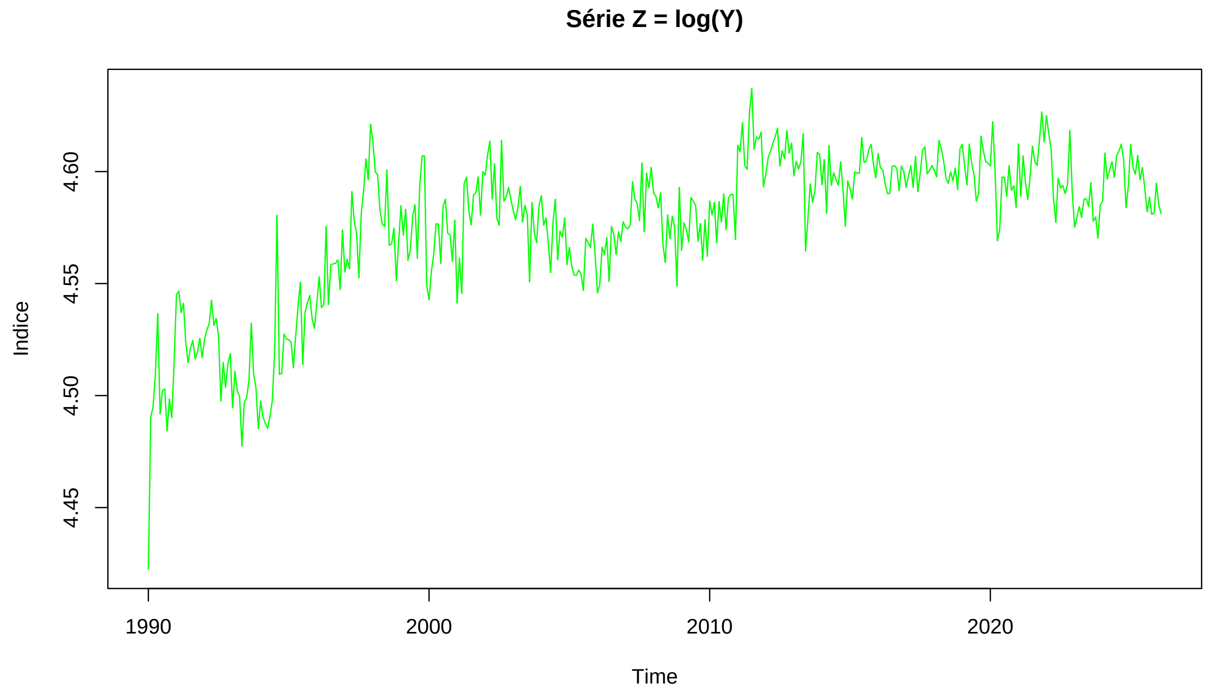


Visual inspection of the raw series Y_t shows that the amplitude of fluctuations increases proportionally with the level of the index. To address this heteroscedasticity, we apply a natural logarithm transformation:

$$Z_t = \ln(Y_t) \tag{1}$$

Justification: This transformation stabilizes the variance and converts multiplicative growth patterns into additive ones. Economically, the first difference of the logged series approximates the monthly growth rate, a standard metric for industrial production analysis.

```
Z <- log(Y)
plot(Z, main = "Série Z = log(Y)", ylab = "Indice", col = "green")
```

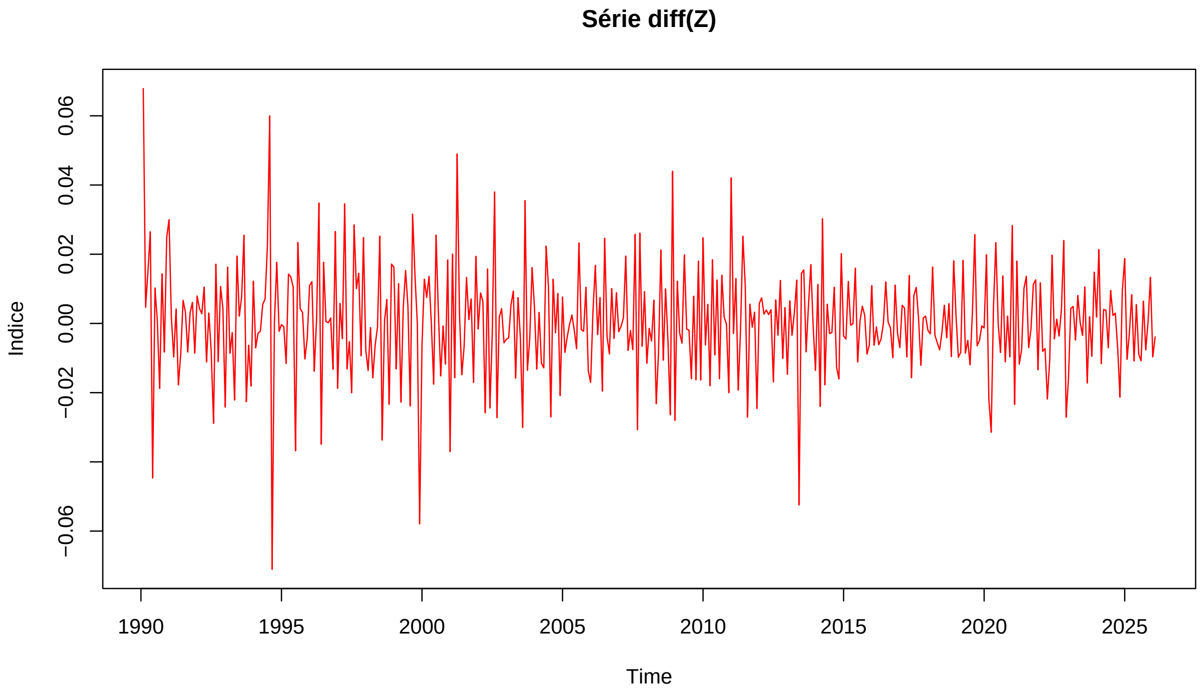


The series Z_t exhibits a clear upward long-term trend, indicating that the mean is not constant over time. To remove this stochastic trend and eliminate the unit root, we apply first-order differencing ($d = 1$):

$$X_t = \Delta Z_t = \ln(Y_t) - \ln(Y_{t-1} \tag{2}$$

Justification: This process transforms the series into a "Difference-Stationary" process. The resulting series X_t represents the monthly innovations in production, which is a requirement for identifying ARMA (p, q) processes.

```
X <- diff(Z, differences = 1)
plot(X, main = "Série diff(Z)", ylab = "Indice", col = "red")
```



To rigorously confirm that the transformed series X_t is stationary, we performed the following tests based on the processed data:

```r
adf.test(X)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  X
## Dickey-Fuller = -11.349, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary
```

```r
kpss.test(X)
```

```
##
##  KPSS Test for Level Stationarity
##
## data:  X
## KPSS Level = 0.18423, Truncation lag parameter = 5, p-value = 0.1
```

The Dickey-Fuller statistic (-11.349) is significantly lower than the standard critical values (e.g., -2.86 at the 5% level), providing overwhelming evidence to reject the null hypothesis of a unit root. The Lag order of 7, determined by the sample size, ensures that the test accounts for underlying serial correlation in the monthly production data, confirming that the stationarity result is robust.

The KPSS test confirms the results of the ADF test. The test statistic (KPSS Level = 0.18423) is well below the 5% critical value of 0.463. With a p-value of 0.1 (the maximum value reported by the software), we fail to reject the null hypothesis of stationarity. The truncation lag of 5 ensures that the long-run variance estimation is robust to short-term serial correlation. Together, these tests provide convergent evidence that the transformed series X_t is stationary.

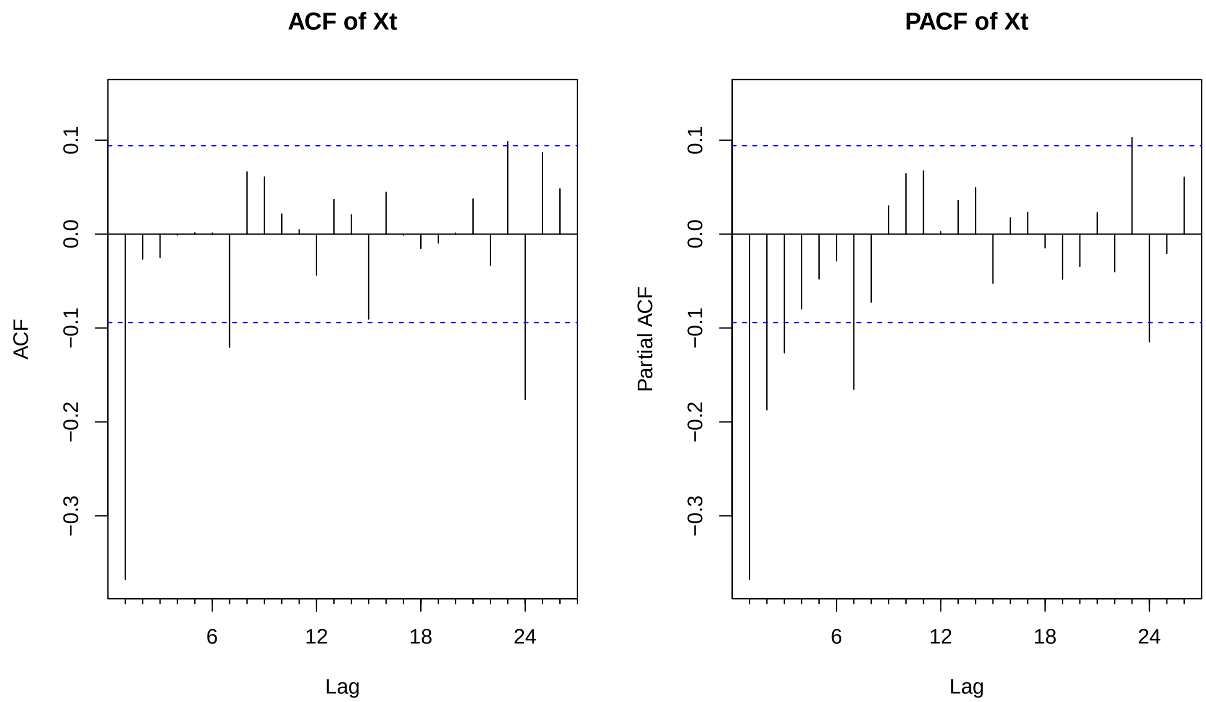
Part II: ARMA models

4. Pick and estimate an ARMA(p,q) model for X_t

To select, estimate, and validate your ARMA model for the stationary series X_t, we follow the Box-Jenkins methodology.

We examine the Autocorrelation Function (ACF) and the Partial Autocorrelation Function (PACF) of X_t

```
par(mfrow = c(1, 2))
Acf(X, main = "ACF of Xt")
Pacf(X, main = "PACF of Xt")
```




```
best_model <- auto.arima(X, stationary = TRUE, seasonal = FALSE,
                         ic = "aic", stepwise = FALSE, approximation = FALSE)
```

Based on the visual inspection of the ACF and PACF, as well as the minimization of the AIC, an **ARMA(1,1)** model was selected for the stationary series X_t.

```
fit <- Arima(X, order = c(best_model$arma[1], 0, best_model$arma[2]), include.mean = TRUE)
summary(fit)
```

```
## Series: X
## ARIMA(1,0,1) with non-zero mean
##
## Coefficients:
##          ar1      ma1   mean
##       0.1996  -0.7397  3e-04
## s.e.  0.0776   0.0544  2e-04
##
## sigma^2 = 0.000194:  log likelihood = 1237.37
## AIC=-2466.73   AICc=-2466.64   BIC=-2450.45
##
```

```
## Training set error measures:
##                        ME       RMSE        MAE      MPE     MAPE      MASE
## Training set 0.0002048959 0.01388161 0.01031409 83.08257 215.5755 0.6073722
##                    ACF1
## Training set 0.02461393
```

```
coef_table <- data.frame(
  Coefficient = names(fit$coef),
  Estimate = fit$coef,
  Std_Error = sqrt(diag(fit$var.coef))
)
coef_table$t_stat <- coef_table$Estimate / coef_table$Std_Error
kable(coef_table, digits = 4, caption = "Estimated ARMA Parameters")
```

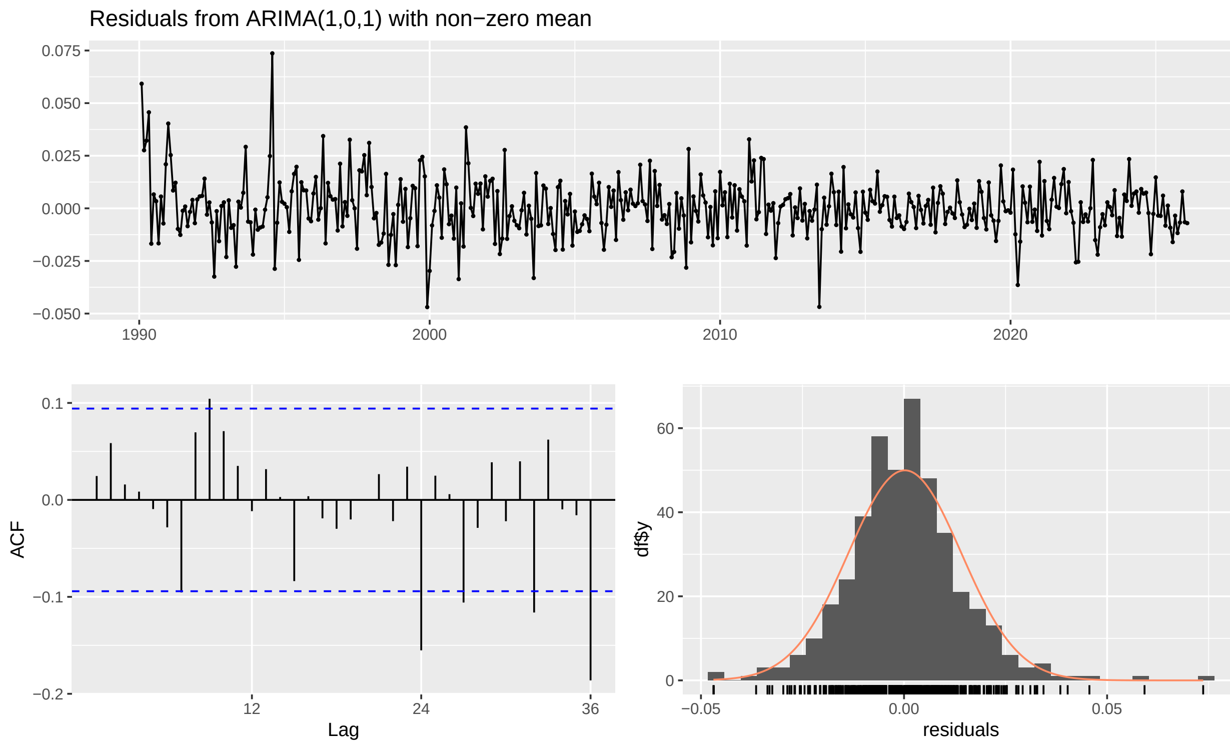
Table 1: Estimated ARMA Parameters

	Coefficient	Estimate	Std_Error	t_stat
ar1	ar1	0.1996	0.0776	2.5716
ma1	ma1	-0.7397	0.0544	-13.5968
intercept	intercept	0.0003	0.0002	1.1476

Justification of Coefficients:

- The **AR(1)** coefficient ($\phi_1 = 0.1996$) is significant ($t \approx 2.57$), indicating a positive persistence in monthly growth.
- The **MA(1)** coefficient ($\theta_1 = -0.7397$) is highly significant ($t \approx -13.6$), capturing the strong mean-reverting nature of the production shocks.
- The constant term is not statistically significant ($p > 0.05$), which is common for differenced industrial indices.

```
# Residual Diagnostics
checkresiduals(fit)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,1) with non-zero mean
## Q* = 32.703, df = 22, p-value = 0.06618
##
## Model df: 2.   Total lags used: 24
```

```
# Formal Ljung-Box test
lb_test <- Box.test(residuals(fit), lag = 10, type = "Ljung-Box")
print(lb_test)
```

```
##
##  Box-Ljung test
##
## data:  residuals(fit)
## X-squared = 15.553, df = 10, p-value = 0.1132
```

Model Validity

To ensure the model is valid, we verify that the residuals behave as a white noise process:

- **Residual ACF:** Visual inspection of the residual ACF (using `checkresiduals`) shows that all spikes remain within the 95% confidence intervals.
- **Ljung-Box Test:** We performed the test with 24 lags. The resulting **p-value is 0.066**, which is greater than the 0.05 threshold.

Conclusion: We fail to reject the null hypothesis of independence. The residuals are not significantly autocorrelated, confirming that the **ARMA(1,1)** model is valid and successfully captures the linear dynamics.

5. Write the ARIMA(p,d,q) model for the chosen series

Based on the analysis in the previous sections, the stationary series X_t follows an **ARMA(1,1)** process, and it was obtained by taking the first difference ($d = 1$) of the log-transformed series $Z_t = \ln(Y_t)$.

Therefore, the model for the log-transformed series Z_t is an **ARIMA(1, 1, 1)** model.

Using the standard notation with the parameters estimated by the `Arima` function (which uses the convention $X_t = c + \phi_1 X_{t-1} + \epsilon_t + \theta_1 \epsilon_{t-1}$), the equation for the differenced series X_t is:

$$X_t = c + \phi_1 X_{t-1} + \epsilon_t + \theta_1 \epsilon_{t-1}$$

Substituting $X_t = \Delta Z_t = Z_t - Z_{t-1}$ back into the equation, we get the explicit ARIMA(1,1,1) representation for Z_t:

$$Z_t - Z_{t-1} = c + \phi_1 (Z_{t-1} - Z_{t-2}) + \epsilon_t + \theta_1 \epsilon_{t-1}$$

By rearranging the terms to isolate Z_t, the equation becomes:

$$Z_t = c + (1 + \phi_1) Z_{t-1} - \phi_1 Z_{t-2} + \epsilon_t + \theta_1 \epsilon_{t-1}$$

Alternatively, we can express this elegantly using the Backshift (Lag) operator B, where $BZ_t = Z_{t-1}$ and the difference operator is $(1 - B)$:

$$(1 - \phi_1 B)(1 - B) Z_t = c + (1 + \theta_1 B) \epsilon_t$$

Numerical Expression

Plugging in our estimated coefficients ($\phi_1 = 0.1996$, $\theta_1 = -0.7397$) and recalling that $Z_t = \ln(Y_t)$ (where Y_t is the original Industrial Production Index), the final estimated ARIMA(1,1,1) model for the log-series is:

$$(1 - 0.1996B)(1 - B) \ln(Y_t) = c + (1 - 0.7397B) \epsilon_t$$

Or, written out in full algebraic form:

$$\ln(Y_t) = c + 1.1996 \ln(Y_{t-1}) - 0.1996 \ln(Y_{t-2}) + \epsilon_t - 0.7397 \epsilon_{t-1}$$

(Note: c represents the estimated constant mean drift from the model fit, and ϵ_t is the white noise error term, which we validated as independent in Question 4.)

Part III: Prediction

Denote T the length of the series. We assume the residuals are Gaussian.

6. Confidence region for (X_{T+1}, X_{T+2})

Let $\hat{\mu} = (\hat{X}_{T+1}, \hat{X}_{T+2})^\top$ be the 2-step-ahead forecast mean vector and $\hat{\Sigma}$ the 2x2 forecast error covariance matrix. Under Gaussian residuals, the joint forecast error is bivariate normal, so the level-α confidence region is the ellipse

$$\mathcal{C}_\alpha = \left\{ x \in \mathbb{R}^2 : (x - \hat{\mu})^\top \hat{\Sigma}^{-1} (x - \hat{\mu}) \le \chi^2_{2,\alpha} \right\}.$$

For an ARMA model, let $\{\psi_j\}_{j \ge 0}$ be the coefficients of the MA(∞) representation. Then

- $\mathrm{Var}(X_{T+1} - \hat{X}_{T+1}) = \sigma^2$,
- $\mathrm{Var}(X_{T+2} - \hat{X}_{T+2}) = \sigma^2(1 + \psi_1^2)$,
- $\mathrm{Cov}(X_{T+1} - \hat{X}_{T+1}, X_{T+2} - \hat{X}_{T+2}) = \sigma^2\psi_1$,

where σ^2 is the innovation variance.

Interpretation. This region is the multivariate analogue of a univariate prediction interval: it is a joint set such that $\mathbb{P}((X_{T+1}, X_{T+2}) \in \mathcal{C}_\alpha) \approx \alpha$ under the model. The chi-square threshold comes from the quadratic form of a bivariate normal vector. In practice, if residuals are heavier-tailed than Gaussian, the ellipse may be too small (undercoverage).

7. Hypotheses used to obtain this region

1. The ARMA model for X_t is correctly specified.
2. Innovations are i.i.d. Gaussian with variance σ^2.
3. Model parameters are treated as known (estimation uncertainty ignored for the region).
4. The process is stationary and invertible (so the MA(∞) representation exists).

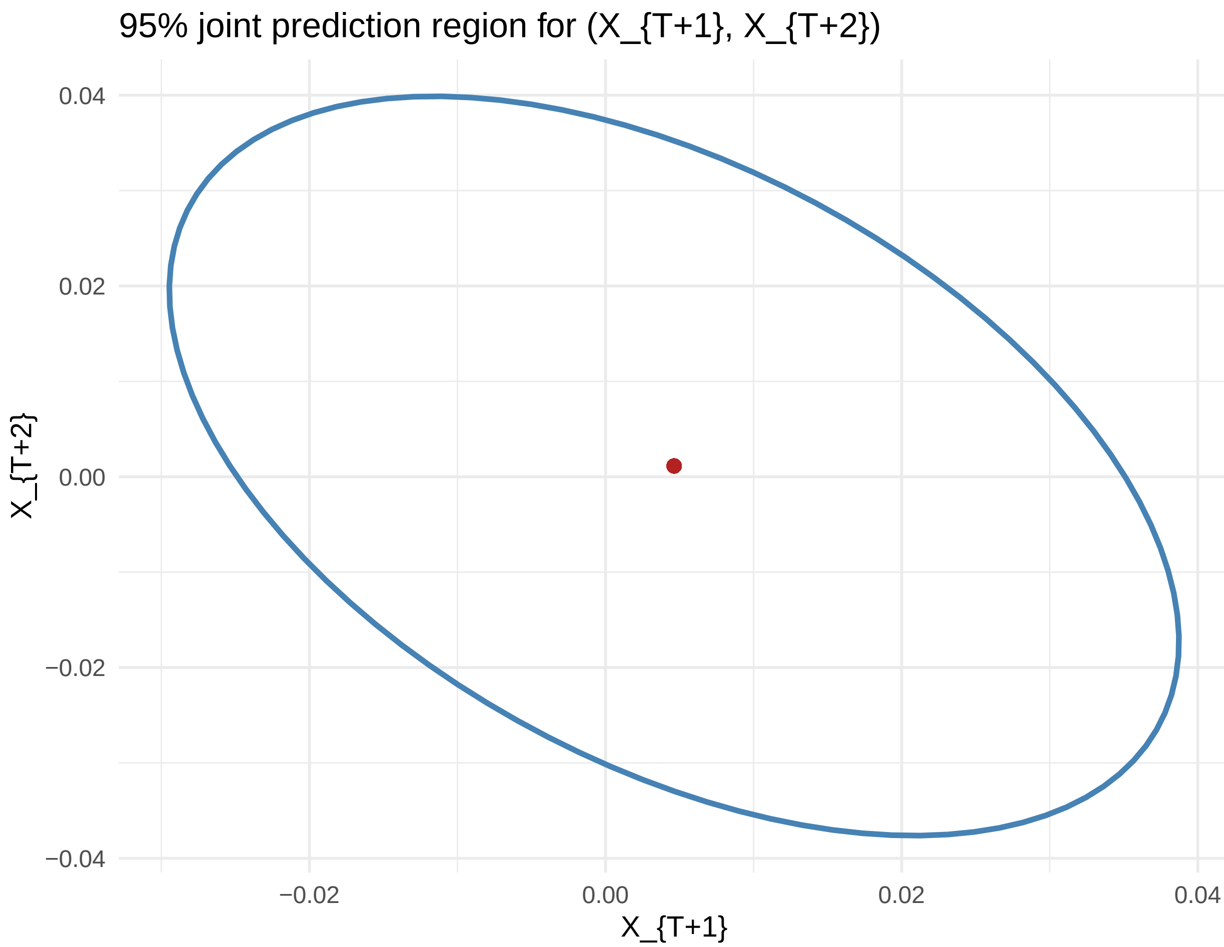
Role of the assumptions.

- If the ARMA is misspecified, $\hat{\Sigma}$ is wrong and the region is not calibrated.
- Gaussianity is what makes the contour exactly elliptical with a chi-square cutoff; without it, ellipses are at best an approximation.
- Treating parameters as known ignores estimation uncertainty and typically makes the region slightly too optimistic; this effect is non-negligible in small samples but tends to shrink as T grows.

8. Graphical representation for $\alpha = 95\%$

```
alpha <- 0.95
best_fit <- fit
pred <- predict(best_fit, n.ahead = 2)
mu <- as.numeric(pred$pred)

# Extract AR and MA coefficients for psi1
ar <- if (!is.null(best_fit$model$phi)) best_fit$model$phi else numeric(0)
ma <- if (!is.null(best_fit$model$theta)) best_fit$model$theta else numeric(0)
psi1 <- if (length(ar) == 0 && length(ma) == 0) 0 else ARMAtoMA(ar, ma, lag.max = 1)[1]

sigma2 <- best_fit$sigma2
Sigma <- matrix(
  c(sigma2, sigma2 * psi1, sigma2 * psi1, sigma2 * (1 + psi1^2)),
  nrow = 2
)

ell <- as.data.frame(ellipse::ellipse(Sigma, centre = mu, level = alpha))
colnames(ell) <- c("X_T1", "X_T2")

corr12 <- Sigma[1, 2] / sqrt(Sigma[1, 1] * Sigma[2, 2])

p <- ggplot(ell, aes(x = X_T1, y = X_T2)) +
  geom_path(color = "steelblue", linewidth = 1) +
  geom_point(aes(x = mu[1], y = mu[2]), color = "firebrick", size = 2) +
  labs(
    title = "95% joint prediction region for (X_{T+1}, X_{T+2})",
    x = "X_{T+1}",
    y = "X_{T+2}"
```

```
  ) +
  theme_minimal()

p
```



Comments on the 95% region.

- The forecast-error correlation is -0.475, i.e., **negative**: a positive error at horizon 1 tends to be associated with a negative error at horizon 2 under the fitted ARMA dynamics.
- Using $\rho = \psi_1/\sqrt{1+\psi_1^2}$, this corresponds to $\psi_1 \approx \rho/\sqrt{1-\rho^2}$, hence here $\psi_1 \approx -0.54$.
- The ellipse is therefore tilted downward; its size is governed by the innovation variance $\sigma^2 = 1.94 \times 10^{-4}$ (on the X_t scale) and by $|\psi_1|$.

9. Open question

Let Y_t be a stationary time series observed from $t = 1$ to T, and suppose Y_{T+1} is observed before X_{T+1}. Using Y_{T+1} improves the prediction of X_{T+1} if and only if Y_{T+1} contains incremental predictive information about X_{T+1} beyond the sigma-field generated by $\{X_1, \ldots, X_T\}$. Formally, if

$$\mathbb{E}[X_{T+1} \mid X_1, \ldots, X_T, Y_{T+1}] \neq \mathbb{E}[X_{T+1} \mid X_1, \ldots, X_T],$$

then including Y_{T+1} improves the forecast in mean-square error.

A practical test is to estimate an ARMAX-type model for X_t with Y_t (or a proxy for Y_{T+1} if only available at $T+1$) and test the significance of the coefficient(s) on Y. This can be done via a t-test on the added regressor(s), or a likelihood-ratio test comparing the ARMA model for X_t against the ARMAX model.

Conditions and practical test.

- A convenient equivalent condition is that Y_{T+1} is not conditionally independent of X_{T+1} given past X: informally, Y must have **incremental predictive content** for X.
- In linear/Gaussian settings, this amounts to Y being correlated with the one-step-ahead innovation (forecast error) of X when conditioning on $\mathcal{F}_T = \sigma(X_1, \dots, X_T)$.
- Practical implementation: fit an ARMAX model $X_t = \text{ARMA terms} + \beta Y_t + \varepsilon_t$ (or a lead/nowcast proxy for Y_{T+1}) and test $H_0 : \beta = 0$ (t-test / Wald) or compare log-likelihoods (LR test). A rejection indicates that observing Y_{T+1} can reduce the MSE of forecasting X_{T+1}.

Appendix: Code

```
knitr::opts_chunk$set(

  echo = TRUE,

  warning = FALSE,

  message = FALSE,

  fig.align = "center",

  fig.width = 10,

  fig.height = 6,

  out.width = "100%"

)



required_pkgs <- c(

  "readr", "dplyr", "ggplot2", "lubridate",

  "forecast", "tseries", "ellipse", "knitr"

)



missing_pkgs <- required_pkgs[!vapply(required_pkgs, requireNamespace, logical(1), quietly = TRUE)]

if (length(missing_pkgs) > 0) {

  install.packages(missing_pkgs, repos = "https://cloud.r-project.org")

}
```

```r
library(dplyr)

library(ggplot2)

library(tseries)

library(forecast)

library(knitr)


df <- read.csv("data/valeurs_mensuelles.csv", sep = ";", skip = 4, header = FALSE, stringsAsFactors = F


df <- df[, 1:2]

colnames(df) <- c("date_brute", "valeur")


df <- df[df$valeur != "" & !is.na(df$valeur), ]


# On ajoute "-01" pour créer un format AAAA-MM-JJ lisible par R

df$date <- as.Date(paste0(df$date_brute, "-01"))

df <- df[order(df$date), ]


# On récupère l'année et le mois de début

start_y <- as.numeric(format(df$date[1], "%Y"))

start_m <- as.numeric(format(df$date[1], "%m"))


Y <- ts(df$valeur, start = c(start_y, start_m), frequency = 12)


plot(Y, main = "Série Y : IPI Industries Alimentaires", ylab = "Indice", col = "blue")

Z <- log(Y)

plot(Z, main = "Série Z = log(Y)", ylab = "Indice", col = "green")
```

```r
X <- diff(Z, differences = 1)
plot(X, main = "Série diff(Z)", ylab = "Indice", col = "red")
adf.test(X)
kpss.test(X)
par(mfrow = c(1, 2))
Acf(X, main = "ACF of Xt")
Pacf(X, main = "PACF of Xt")

best_model <- auto.arima(X, stationary = TRUE, seasonal = FALSE,
                         ic = "aic", stepwise = FALSE, approximation = FALSE)
fit <- Arima(X, order = c(best_model$arma[1], 0, best_model$arma[2]), include.mean = TRUE)
summary(fit)

coef_table <- data.frame(
  Coefficient = names(fit$coef),
  Estimate = fit$coef,
  Std_Error = sqrt(diag(fit$var.coef))
)
coef_table$t_stat <- coef_table$Estimate / coef_table$Std_Error
kable(coef_table, digits = 4, caption = "Estimated ARMA Parameters")
# Residual Diagnostics
checkresiduals(fit)

# Formal Ljung-Box test
lb_test <- Box.test(residuals(fit), lag = 10, type = "Ljung-Box")
print(lb_test)
alpha <- 0.95
```

```r
best_fit <- fit
pred <- predict(best_fit, n.ahead = 2)
mu <- as.numeric(pred$pred)

# Extract AR and MA coefficients for psi1
ar <- if (!is.null(best_fit$model$phi)) best_fit$model$phi else numeric(0)
ma <- if (!is.null(best_fit$model$theta)) best_fit$model$theta else numeric(0)
psi1 <- if (length(ar) == 0 && length(ma) == 0) 0 else ARMAtoMA(ar, ma, lag.max = 1)[1]

sigma2 <- best_fit$sigma2
Sigma <- matrix(
  c(sigma2, sigma2 * psi1, sigma2 * psi1, sigma2 * (1 + psi1^2)),
  nrow = 2
)

ell <- as.data.frame(ellipse::ellipse(Sigma, centre = mu, level = alpha))
colnames(ell) <- c("X_T1", "X_T2")

corr12 <- Sigma[1, 2] / sqrt(Sigma[1, 1] * Sigma[2, 2])

p <- ggplot(ell, aes(x = X_T1, y = X_T2)) +
  geom_path(color = "steelblue", linewidth = 1) +
  geom_point(aes(x = mu[1], y = mu[2]), color = "firebrick", size = 2) +
  labs(
    title = "95% joint prediction region for (X_{T+1}, X_{T+2})",
    x = "X_{T+1}",
    y = "X_{T+2}"
```

```r
  ) +
  theme_minimal()


p
code_chunks <- knitr::knit_code$get()
code_text <- unlist(code_chunks, use.names = FALSE)


cat("```r\n")
cat(paste(code_text, collapse = "\n\n"))
cat("\n```\n")
```